

The Linear Probability Model, Logit, and Probit

Eight true stories behind one S-shaped curve — and hands-on in Stata

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Course: MICROECONOMETRICS

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Your life is a sequence of zeros and ones

Many of the most important things in life are not “how much” but “**yes or no**”:

- Did you pass the university entrance exam?
- Did you get the job?
- Was your loan approved?
- Do you smoke?
- Did you vote in the last election?
- Did you take the train or drive?
- Did the rocket explode?
- Did it rain?

The question for today

How do we model the **probability** that something happens ($Y = 1$) as a function of characteristics (X)? And why is ordinary OLS *almost* right, but not quite?

Every example today is a real case. This model has decided matters of life and death, won a Nobel Prize, and been argued before the Supreme Court.

Road map: eight stories, one model

Story	Year	What it teaches
<i>Part I — the stories and the theory</i>		
1 Bliss and the grape-leaf insects	1934	Why an S-curve; the birth of the probit
2 “70% chance of rain”	1964	The linear probability model (LPM) and its flaws
3 The space shuttle <i>Challenger</i>	1986	Estimating a logit ; sample selection; extrapolation
4 Elo chess ratings	1960	Probit versus logit: when does it matter?
5 McFadden and the BART trains	1972	From utility to logit; prediction; a Nobel Prize
6 <i>McCleskey v. Kemp</i>	1987	Reading an odds ratio ; the limits of interpretation
7 Spain’s VioGén risk score	2007–	Classification errors with lethal cost
8 Austria’s job-seeker algorithm	2018	A coefficient on a group dummy, and a cutoff
<i>Part II — hands-on in Stata with <code>auto.dta</code></i>		

Stories are in teaching order, not date order. Part II then works through a complete Stata example: estimation, odds ratios, margins, classification, ROC.

Notation: a binary dependent variable

- $Y_i \in \{0, 1\}$; for example $Y_i = 1$ if household i is poor.
- X_i : characteristics (education, income, temperature, dose, ...).
- Because Y is only ever 0 or 1, **its expected value is a probability**:

$$E[Y_i | X_i] = 1 \cdot P(Y_i = 1 | X_i) + 0 \cdot P(Y_i = 0 | X_i) = P(Y_i = 1 | X_i).$$

- So “modelling Y ” means choosing a function F such that

$$P(Y_i = 1 | X_i) = F(\beta_0 + \beta_1 X_i).$$

The three choices of F we will meet

LPM	$F(z) = z$	a straight line
Probit	$F(z) = \Phi(z)$	the standard normal CDF
Logit	$F(z) = \Lambda(z) = \frac{e^z}{1 + e^z}$	the logistic CDF

Story 1 — An entomologist loses his job (1933)

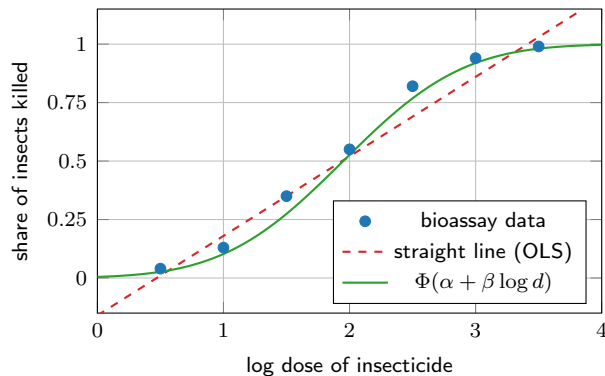


R. A. Fisher (1890–1962), under whom Bliss worked out the probit in London. No free photograph of Bliss himself survives. Photo: public domain (1913).

- **Chester Bliss** is an entomologist at the US Department of Agriculture. In 1933 the Depression cuts the budget and he is laid off.
- He sails to London and spends two years studying under **R. A. Fisher**.
- The problem he brings with him: how to compare insecticides. The data always form an *S-shaped curve*: a small dose kills few insects, a large dose kills almost all of them.
- Regression, at the time, is for straight lines only. His fix (1934): convert “percent killed” into a ***probability unit*** — a **probit**.
- He goes on to the Institute of Plant Protection in Leningrad (1936–38), and finally to Connecticut.

To this day, every LD_{50} printed on a pesticide or a drug is computed with Bliss’s probit.

Story 1 — What Bliss was looking at: an S-curve



Two things wrong with the straight line:

- 1 At low doses it predicts a **negative** share; at high doses a share **above one**.
- 2 The data bend: the effect of dose is large in the middle and small at the ends.

Bliss's idea: suppose each insect has a *tolerance threshold* T_i , normally distributed across insects. The insect dies when the dose exceeds its tolerance:

$$P(\text{dies} \mid d) = P(T_i \leq \log d) = \Phi(\alpha + \beta \log d).$$

This is the **probit model**: an S-curve born from a latent variable.

Story 2 — Where does “70% chance of rain” come from? (1964)

KSFO	GFS MOS GUIDANCE											9/24/2026 1200 UTC										
DT /SEPT 24/SEPT 25												/SEPT 26										
HR	18	21	00	03	06	09	12	15	18	21	00	03	06	09	12	15	18	21	00	06	12	
N/X							61					72				55			72		56	
TMP	67	73	71	66	65	63	62	62	68	72	67	61	59	58	57	58	66	70	67	59	57	
DPT	58	56	53	51	51	54	55	55	54	52	51	51	51	50	51	52	51	51	51	51	51	
CLD	FW	SC	FW	FW	FW	CL	SC	BK	SC	FW	SC	FW	FW	CL	FW	BK	BK	SC	BK	SC	BK	
WDR	32	30	28	29	28	27	27	27	28	28	27	28	27	27	27	27	29	28	27	26	26	
WSP	10	15	19	15	13	11	11	09	12	18	21	18	12	11	10	08	10	13	16	12	12	
P06			0		2		0		0		1		0		0		0		0	1	1	
P12							2				1				0			2		1		
Q06			0		0		0		0		0		0		0		0		0	0	0	
Q12							0				0				0				0	0	0	
T06		0/	0/	0/	0/	0/	0/	0/	0/	0/	0/	0/	0/	0/	0/	1/	0/	1/	0/	0/	0/	
T12			3/	2/			0/	1/			1/	0/			1/	0/		4/	0/			
POZ	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
POS	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
TYP	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
CIG	8	8	8	8	8	8	8	4	8	8	8	8	8	8	8	5	8	8	8	8	8	
VIS	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	
OBV	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	

A live GFS MOS bulletin for San Francisco (KSFO), 24 September 2026. The boxed P12 row is the 12-hour probability of precipitation, in percent. Source: NOAA/NWS (public domain).

- 1964: **Robert Miller** of the US Weather Bureau proposes **REEP** (*Regression Estimation of Event Probabilities*): regress a rain indicator $Y \in \{0, 1\}$ on the output of the numerical weather model using **plain OLS**. Call the fitted value a “probability”.
- 1965: the US begins issuing *probability of precipitation* forecasts to the public — the first country to do so.
- 1972: Glahn and Lowry build the **MOS** system on top of it. The National Weather Service runs it to this day.
- What if a prediction comes out below 0 or above 1? It is **clipped** to 0 or 1. That’s it.

What this means

The rain probability on your phone is a direct descendant of the **linear probability model (LPM)**.

The Linear Probability Model (LPM)

Model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \text{so that} \quad P(Y_i = 1 \mid X_i) = \beta_0 + \beta_1 X_i.$$

Estimated by ordinary OLS: regress y on x

Strengths

- β_1 reads directly: *a one-unit increase in X raises the probability that $Y = 1$ by β_1 .*
- Simple and fast; in a randomised experiment the coefficient is the causal effect on the probability.
- Widely used in applied economics; Angrist and Pischke (2009) argue its AMEs are usually close to those of a probit.

Weaknesses

- Predictions can fall below 0 or above 1 — which is not a probability.
- The marginal effect is constant: one extra degree has the same effect at 30° as at 80° . Does that make sense?
- The error u_i takes only two values, so it is **heteroskedastic**: $\text{Var}(u_i \mid X_i) = p_i(1 - p_i)$. Use `robust`.

Story 3 — The night before the launch (27 January 1986)



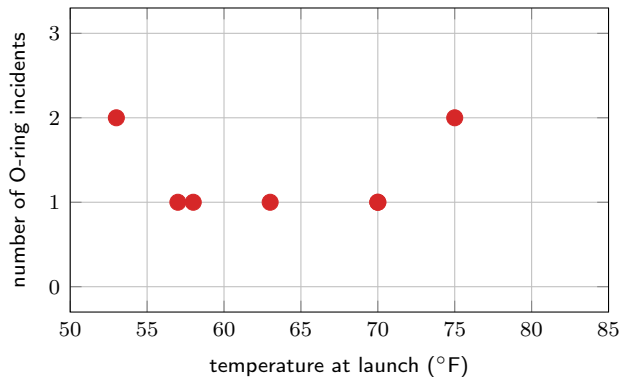
Icicles on the launch tower at Pad 39B on the morning of 28 January 1986, with the solid rocket booster behind. Photo: NASA (public domain).

- The forecast for the morning: about -2°C ($\approx 31^{\circ}\text{F}$). No shuttle has ever launched below 53°F .
- Engineers at Morton Thiokol, on a teleconference, **argue against** launching: the rubber **O-rings** that seal the booster joints stiffen in the cold.
- They show a chart of O-ring damage against temperature for the flights that *had* damage. NASA's response: "There is no pattern." Thiokol's management reverses itself and approves the launch.
- 11:38 a.m.: lift-off. 73 seconds later *Challenger* breaks apart. All seven crew members are killed.

We are going to look at the chart they saw — and at the chart they *should* have seen.

Story 3 — The chart the engineers saw that night

Only the 7 flights that had damage



What do you see?

- Damage at 53°F *and* at 75°F.
- At first glance: “temperature doesn’t matter”.

What is missing from this chart?

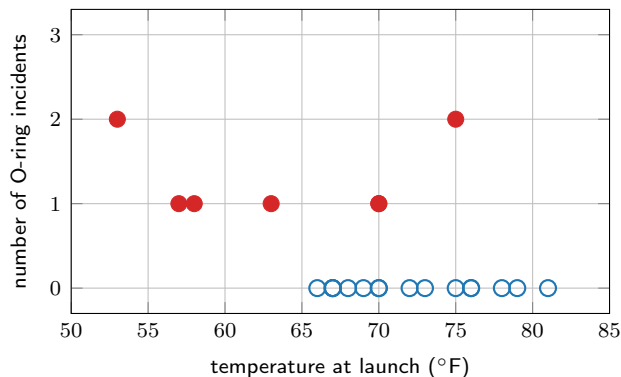
- The **other 16 flights**, which had *no* damage — all dropped because “there was nothing to plot”.

A classic mistake

Selecting the sample on the **value of the dependent variable**. The signal was in the zeros that were thrown away.

Story 3 — The full picture: all 23 flights

All 23 flights before *Challenger*



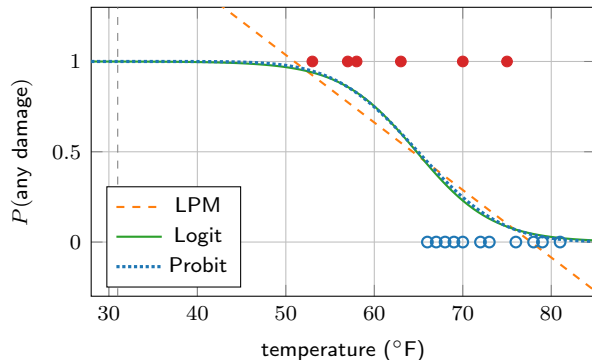
Now the pattern is obvious:

- Every flight below 65°F had damage (4 out of 4).
- Above 65°F: only 3 out of 19.

And *Challenger* was launched at 31°F — 22° colder than anything ever attempted.

Data: Dalal, Fowlkes and Hoadley (1989), *JASA*. Available as `orings` in R (package `faraway`).

Story 3 — Three models, one question: what is the probability of damage at 31°F?



	LPM	Logit	Probit
constant	2.905	15.04	8.77
temperature	-0.037	-0.232	-0.135
$\hat{P}$ at 81°F	-0.12	0.02	0.02
$\hat{P}$ at 70°F	0.29	0.23	0.25
$\hat{P}$ at 53°F	0.92	0.94	0.95
$\hat{P}$ at 31°F	1.75	0.9996	> 0.9999

- LPM: “a 175% probability” — impossible, but the direction is right.
- Logit and probit: damage all but certain.
- Dalal et al.’s full risk analysis:
 $P(\text{catastrophe}) \geq 13\%$.

Logit and probit: the functional forms

Logit

$$P(Y = 1 | X) = \Lambda(\beta_0 + \beta_1 X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Equivalent to a linear model for the **log-odds**:

$$\ln \frac{P}{1 - P} = \beta_0 + \beta_1 X$$

Probit

$$P(Y = 1 | X) = \Phi(\beta_0 + \beta_1 X) = \int_{-\infty}^{\beta_0 + \beta_1 X} \phi(z) dz$$

Equivalent to a **latent variable** model (remember Bliss):

$$Y^* = \beta_0 + \beta_1 X + \varepsilon, \quad Y = \mathbf{1}[Y^* > 0], \quad \varepsilon \sim N(0, 1)$$

What they share

- Always $0 < P < 1$; an S-curve; the effect of X is largest near $P \approx 0.5$ and fades at the tails.
- Estimated by *maximum likelihood*: find the β that makes the observed zeros and ones “most likely”:
 $\ln L(\beta) = \sum_i [Y_i \ln F(X_i \beta) + (1 - Y_i) \ln(1 - F(X_i \beta))]$.
- The coefficient β_1 is **not** a marginal effect (more on this later). Its sign and significance can still be read directly.

Story 3 — Estimation in Stata (23 rows of data — it fits on one screen)

Enter the data (paste as is)

```
clear
set obs 23
local t "66 70 69 68 67 72 73 70 57 63 70 78"
local t "`t' 67 53 67 75 70 81 76 79 75 76 58"
local d "0 1 0 0 0 0 0 0 1 1 1 0 0 1 0 0"
local d "`d' 0 0 0 0 1 0 1"
generate temp = real(word("`t'", _n))
generate damage = real(word("`d'", _n))
tab damage
```

Why not input?

input reads **one observation per line**. Put four pairs on a line and Stata silently keeps only the first: you get 6 flights, not 23.

Three models, then margins

```
regress damage temp, robust
margins, at(temp=(31 53 70 81))

logit damage temp
margins, at(temp=(31 53 70 81))
margins, dydx(temp)

probit damage temp
margins, dydx(temp)
```

Check your output: 23 observations; the coefficients and $\hat{P}$ in the table on the previous slide; average marginal effects -0.033 (logit), -0.034 (probit) against the LPM's -0.037 .

The full script is `stata/challenger.do`.

Story 3 — Epilogue: ice, rubber, and three lessons



The crew of STS-51-L. Back row: Onizuka, McAuliffe, Jarvis, Resnik. Front row: Smith, Scobee, McNair.
Photo: NASA (public domain).

- 1 **Never select the sample on Y .** Seven “damaged” points without the sixteen “clean” ones erase all the information about temperature.
- 2 **Fit a curve, not a line.** The LPM gets the direction right but produces impossible numbers; logit and probit give probabilities you can take into a meeting room.
- 3 **Extrapolation is dangerous — but refusing to extrapolate is more dangerous.** 31°F is far outside the data; every model says the risk is extreme.

Dalal, Fowlkes and Hoadley (1989), “Risk Analysis of the Space Shuttle: Pre-Challenger Prediction of Failure”, *JASA* 84(408): 945–957.

Story 4 — A physics professor who was tired of chess rankings (1960)



Arpad Elo (1903–1992). Photo: Milwaukee Sentinel, via Wikipedia; low-resolution copy used under fair use for teaching.

- **Arpad Elo**: professor of physics at Marquette University and eight-time Wisconsin chess champion.
- His model: a player's performance in any game is a **normal** random variable centred on their true strength R . Hence

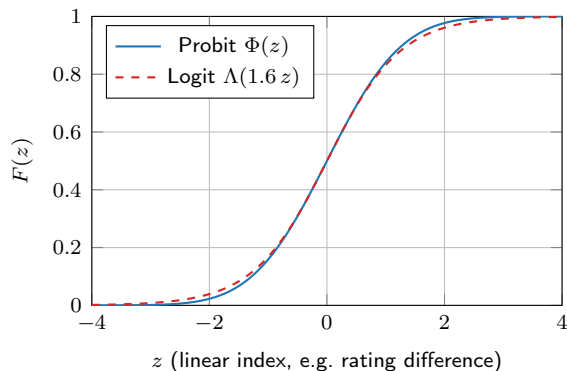
$$P(A \text{ wins}) = \Phi\left(\frac{R_A - R_B}{\sigma\sqrt{2}}\right) \quad \text{— a **probit**.}$$

- Adopted by the US Chess Federation (1960) and by FIDE (1970).
- The USCF then noticed that weaker players *win more often* than the normal curve predicts — and switched to the **logistic** curve:

$$P(A \text{ wins}) = \frac{1}{1 + 10^{-(R_A - R_B)/400}} \quad \text{— a **logit**.}$$

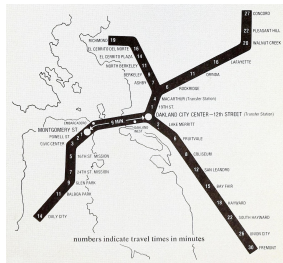
- FIDE kept the normal. Two federations, the same data, probit versus logit.

Probit versus logit: when does the difference show?



- Once rescaled, the two curves **almost coincide**. The largest gap is below 0.02.
- The logistic distribution has **fatter tails**: “surprises” (the weaker player winning) are slightly more likely. That is exactly what the USCF saw.
- Rule of thumb: $\hat{\beta}_{\text{logit}} \approx 1.6\text{--}1.7 \times \hat{\beta}_{\text{probit}}$.
Challenger: $-0.232 / -0.135 = 1.72$.
- Marginal effects and predicted probabilities: practically identical. **Choose by the convention of your field**, not by the results.

Story 5 — The forecast that won a Nobel Prize (1972)



Left: BART's 1972 travel-time map (BART District, public domain). Right: Daniel McFadden. Photo: Bengt Oberger, CC BY-SA 4.0.

- San Francisco is building a new rapid-transit system, **BART**. The official forecast: **15%** of commuters will switch to it.
- **Daniel McFadden**, a young economist at Berkeley, surveys commuters *before* BART opens: which mode they use, how long it takes, what it costs.
- His model: each person chooses the mode with the **highest utility**; utility is a systematic part plus a random shock. The result is a **logit**.
- McFadden's forecast: **6.3%**. The actual share: **6.2%**.
- 2000: the Nobel Prize in Economics, "for his development of theory and methods for analyzing discrete choice".

From utility to logit: why economists love this model

Suppose individual i chooses between **BART** ($Y = 1$) and **the car** ($Y = 0$).

$$U_{i1} = X_i\beta_1 + \varepsilon_{i1}, \quad U_{i0} = X_i\beta_0 + \varepsilon_{i0}$$

She takes BART if $U_{i1} > U_{i0}$:

$$P(Y_i = 1 \mid X_i) = P(\varepsilon_{i0} - \varepsilon_{i1} < X_i(\beta_1 - \beta_0)) = F(X_i\beta)$$

If ε is type-I *extreme value*

The difference is **logistic** $\Rightarrow$ **logit**.

A closed form, easy to compute, and it extends to $J > 2$ alternatives (mlogit, next lecture).

If ε is *normal*

The difference is **normal** $\Rightarrow$ **probit**.

Statistically more “natural”, but with many alternatives it needs high-dimensional integrals.

The heart of McFadden's contribution

The logit is not just a trick for drawing S-curves. It **follows from utility-maximising behaviour** — so its coefficients can be read as preferences, and the model can forecast policies that have never existed.

Story 6 — A regression before the Supreme Court (1987)



The Supreme Court of the United States. Photo: Joe Ravi, CC BY-SA 3.0.

- Atlanta, 1978. **Warren McCleskey**, a Black man, is sentenced to death for killing a white police officer during a robbery.
- His lawyers present the **Baldus study**: more than 2,000 murder cases in Georgia, a **logit** model with 39 control variables (severity of the crime, criminal record, and so on).
- The finding: defendants who killed **white** victims were sentenced to death with an **odds ratio of 4.3** relative to those who killed Black victims.
- The ruling, 5–4: the statistics are sound, but they do not prove discrimination *in McCleskey's own case*. He is executed in 1991. Justice Powell later calls it the one vote he would change.

Reading logit coefficients: odds and odds ratios

Definitions

$$\text{odds} = \frac{P}{1-P}, \quad \ln(\text{odds}) = X\beta$$

A one-unit increase in X_k multiplies the odds by e^{β_k} , the **odds ratio**.

The Baldus study

$\beta_{\text{white victim}} \approx \ln 4.3 \approx 1.46$.

The odds of a death sentence are 4.3× higher, *ceteris paribus*.

In Stata: `logit y x`, or `logistic y x`

What you can and cannot say

- An odds ratio is not a ratio of probabilities. When P is small they are close; when P is large they differ a lot.
- A coefficient describes a **pattern in the population**, not the motive behind **one** decision.
- Controlling for 39 variables is not controlling for everything: an “omitted variable” objection is always available.
- Statistically significant is not legally “proven”. The Court accepted the numbers and rejected the conclusion.

Marginal effects: what does β mean in a logit or probit?

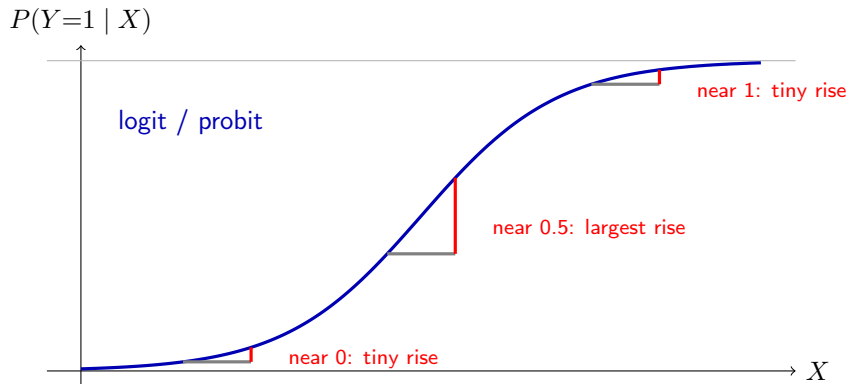
$$\frac{\partial P(Y = 1 | X)}{\partial X_k} = f(X\beta) \beta_k, \quad f = \text{the pdf: } \lambda(z) = \Lambda(z)[1 - \Lambda(z)] \text{ (logit), } \phi(z) \text{ (probit)}$$

- The marginal effect **depends on** X : large in the middle of the curve ($P \approx 0.5$), small at the tails.
- Challenger: the effect of one extra degree is ≈ -0.041 at 70°F , ≈ -0.013 at 53°F , and ≈ -0.005 at 81°F .
- Two summaries are commonly reported:
 - **AME** (*average marginal effect*): the average of the effect over all observations $\rightarrow$ margins, `dydx(*)`
 - **MEM** (*marginal effect at the mean*): the effect evaluated at $\bar{X}$ $\rightarrow$ margins, `dydx(*) atmeans`
- For a *dummy*: report $F(X\beta | d=1) - F(X\beta | d=0)$; Stata does this if you write `i.d.`

A fair comparison

Compare the logit **AME**, the probit AME and the LPM coefficient: same scale (a change in probability), usually **very close**. Never compare raw coefficients.

Why the effect is not constant: one picture



Grey: the same step in X at three places. **Red:** the rise in P that step causes.

Intuition: the same step in X moves the probability a lot around $P \approx 0.5$ and hardly at all near 0 or 1. A car at 4,000 lbs is already almost surely domestic; another 100 lbs cannot make it more so. That is why one number (β) cannot be “the effect”, and why we average: the AME.

Summary: LPM versus logit versus probit

	LPM	Logit	Probit
$P(Y = 1 X)$	$X\beta$	$\Lambda(X\beta)$	$\Phi(X\beta)$
Estimation	OLS	ML	ML
Predictions inside $[0, 1]$?	no	yes	yes
Marginal effect	constant = β_k	$\lambda(X\beta)\beta_k$	$\phi(X\beta)\beta_k$
Direct interpretation	change in probability	log-odds; $e^\beta = \text{odds ratio}$	latent index (in s.d.)
Heteroskedastic?	yes (use robust)	not an issue	not an issue
Stata	regress	logit, logistic	probit
Stories	weather forecasts (MOS)	Challenger, BART, Baldus, AMS, VioGén	Bliss, Elo (FIDE)

Practical advice: to learn the direction and significance of an effect, an LPM with robust is enough and easy to read. For **predicted probabilities**, and for effects that are non-linear at the tails of the distribution, use a logit or probit and report margins.

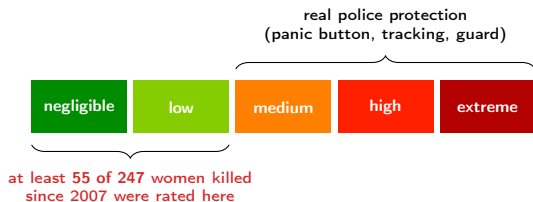
Story 7 — Spain's VioGén: a risk score for women who report abuse (2007–)



Headline card made for this deck. The QR code opens the article:
A. Satariano and R. Toll Pifarré, *The New York Times*, 18 July 2024.

- Since 2007, when a woman reports abuse to the police, an officer fills in a questionnaire (35–39 items): how severe past assaults were, the aggressor, her vulnerability, aggravating factors.
- The score predicts a **new assault**. Its items were chosen and weighted using **odds ratios from logistic regression**; the exact formula has never been published. Reported AUC: 0.66–0.80.
- Five levels: *negligible, low, medium, high, extreme*. The higher levels bring a panic button, tracking of the aggressor, or police guarding the home.
- About 600,000 cases had gone through VioGén by 2020.

Story 7 — When the classifier is wrong, someone can die



95% of scores kept as the software set them
3% of victims rated *medium* or higher

Sources: Eticas external audit (2022); *The New York Times* (2024).

Lobna Hemid. On 11 January 2022 her husband attacked her at home near Madrid. She reported it; VioGén rated her *low risk*. Seven weeks later he stabbed her to death.

January 2025: Spain launched VioGén 2, which drops the lowest (“no risk”) level.

The lessons (measured in Part II, §6)

- A false negative here is not a statistic.
- Where the cutoff for protection sits decides who is protected.
- AUC 0.7 ranks cases better than chance — not well enough for these decisions.
- Officers rarely overrode the score: *automation bias*.

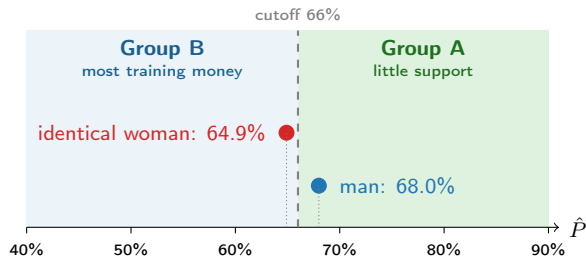
Story 8 — An algorithm with a coefficient for being female (Austria, 2018)



An office of the AMS, Austria's public employment service, in Villach. Photo: Naturpuur, CC BY-SA 4.0.

- The **AMS** wanted its training budget to go where it helps most, so it had job seekers scored by their chance of finding work.
- $Y = 1$: at least 90 days of unsubsidised work within 7 months (a second model: 180 days within 24 months).
- X : gender, age, citizenship, education, health impairment, care obligations (*recorded for women only*), sector, region, past employment.
- The model: **logistic regression** — 96 of them, by outcome, group of job seeker and stage of unemployment. Reported precision 80–91%.
- The decision:
 - Group **A**, $\hat{P}(\text{short-term job}) \geq 66\%$: little support.
 - Group **C**, $\hat{P}(\text{long-term job}) \leq 25\%$: little support, referred elsewhere.
 - Group **B**, everyone else: most of the training money.
- Tested from 2018; banned by the Data Protection Authority in 2020. A court sided with AMS in 2025 — by then the project was dead.

Story 8 — One coefficient, one cutoff



Same age, education, history and region. The only difference is the gender dummy.

The arithmetic

- Coefficient on female: $\hat{\beta} = -0.14$. Odds ratio $e^{-0.14} = 0.87$: a woman's odds of finding work are **13% lower**.
- A man at $\hat{P} = 0.68$: log-odds $\ln(0.68/0.32) = 0.754$. The same woman: $0.754 - 0.14 = 0.614 \Rightarrow \hat{P} = 0.649$.
- Near the cutoff:
 $\hat{P}(1 - \hat{P})\hat{\beta} \approx 0.66 \times 0.34 \times (-0.14) = -3.1$ points — enough to change her group.

The critics' point

The coefficient accurately describes past outcomes — including past discrimination. Using it to allocate training turns yesterday's disadvantage into tomorrow's.

Part II — From stories to practice: the `auto.dta` example

The data: Stata's 1978 automobile data, 74 cars.

- $Y = \text{foreign}$: 1 if the car is imported (22 cars, 29.7%), 0 if domestic (52 cars).
- $X = \text{weight}$ (pounds) and mpg (miles per gallon).
- Question: what kind of car is likely to be an import?

Stata

```
sysuse auto, clear  
tab foreign
```

Every number and graph in this part comes from `stata/auto_logit.do`. Run it yourself.

What we will do, in order:

- 1 Estimate the logit and read the output
- 2 Odds ratios
- 3 Predicted probabilities
- 4 Marginal effects: AME, MEM, `margins at()`, `marginsplot`
- 5 Interactions and factor variables
- 6 How good is the model? Classification, ROC, calibration
- 7 When things go wrong
- 8 Standard errors: robust, clustered, survey

1. Estimating the logit

```
. logit foreign weight mpg
```

```
Logistic regression           Number of obs =      74
                              LR chi2(2)      =   35.72
                              Prob > chi2     =  0.0000
Log likelihood = -27.175156    Pseudo R2     =  0.3966
```

	foreign	Coefficient	Std. err.	z	P> z
weight		-.0039067	.0010116	-3.86	0.000
mpg		-.1685869	.0919175	-1.83	0.067
_cons		13.70837	4.518709	3.03	0.002

Same model, other estimators: probit: -0.0023 , -0.104
(ratio to logit ≈ 1.65). regress: -0.00047 , -0.019 .

What the coefficients say

- $\text{weight} < 0$: heavier cars are less likely to be imports (in 1978, imports were small).
- $\text{mpg} < 0$ *given weight*: among cars of the same weight, the thriftier ones lean domestic. Only marginally significant ($p = 0.067$).
- Signs and p -values can be read directly. The **sizes cannot**: these are log-odds, not probabilities.

1. Reading the output

Item	Meaning
Coefficient	Change in the log-odds for a one-unit change in X
Std. err.	Sampling variability of the estimate
$z, P > z $	Test of $H_0: \beta_k = 0$ (the Wald test)
Log likelihood	$\ln L$ at the maximum: -27.18 here; -45.03 for a model with only a constant
LR chi2(2)	$2 [\ln L - \ln L_0] = 2 [-27.18 - (-45.03)] = 35.72$: do the regressors improve the fit at all?
Pseudo R^2	McFadden's $1 - \ln L / \ln L_0 = 1 - 27.18/45.03 = 0.397$

- The LR test plays the role of the F -test in OLS: it compares two *nested* models by their log likelihoods.
- Pseudo- R^2 is **not** a share of variance explained. Use it only to compare models on the same data; 0.2–0.4 already counts as a good fit for a logit.
- Above the table Stata prints the iteration log of the ML search. If it does not converge, do not trust the numbers.

2. Odds ratios: logistic

```
. logistic foreign weight mpg
```

	foreign	Odds ratio	Std. err.	z	P> z
weight		.9961009	.0010077	-3.86	0.000
mpg		.8448578	.0776572	-1.83	0.067
_cons		898396.7	4059594	3.03	0.002

Note: _cons estimates baseline odds.

Same model, same z and p ; only the column differs: $e^{\hat{\beta}}$ instead of $\hat{\beta}$. `logit ...`, or does the same.

Two traps

Odds are not probabilities: $OR = 2$ does not mean “twice as likely”. And the OR depends on the unit of X : rescale before you interpret.

How to read an odds ratio

- $OR = 1$: no effect. $OR = 2$: the odds double. $OR = 0.8$: the odds fall by 20%.
- `mpg`: $OR = 0.845$. One extra mile per gallon cuts the odds of being an import by about 15%.
- `weight`: $OR = 0.9961$ *per pound* — looks like nothing. Per 100 pounds: $0.9961^{100} = e^{100\hat{\beta}} = 0.68$, so the odds fall by a third.

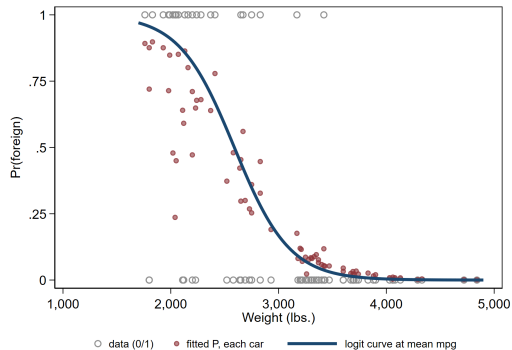
3. Predicted probabilities

```
. predict phat
(option pr assumed; Pr(foreign))
```

```
. list make foreign weight mpg phat in 1/8
```

make	foreign	weight	mpg	phat
AMC Concord	Domestic	2,930	22	.19044
AMC Pacer	Domestic	3,350	17	.09578
AMC Spirit	Domestic	2,640	22	.42208
Buick Century	Domestic	3,250	20	.08626
Buick Electra	Domestic	4,080	15	.00849
Buick LeSabre	Domestic	3,670	18	.02499
Buick Opel	Domestic	2,230	26	.64866
Buick Regal	Domestic	3,280	20	.07746

phat is $\hat{P}(Y = 1 | X_i) = \Lambda(X_i\hat{\beta})$ for every car. The Buick Opel, light and frugal, is “predicted” to be an import (0.65). It was in fact built by Opel in Germany and sold under a Buick badge.



Grey circles: the 0/1 data. Red dots: phat for each car — scattered because each car has its own mpg. Blue curve: the logit S-curve with mpg held at its mean (21.3).

4. Marginal effects: AME and MEM

$$\text{AME} = \frac{1}{N} \sum_{i=1}^N f(X_i \hat{\beta}) \hat{\beta}_k \quad \text{MEM} = f(\bar{X} \hat{\beta}) \hat{\beta}_k$$

- **AME** averages the effect over the actual sample. Report this one.
- **MEM** evaluates it at the “average car”, which may not exist.
- Standard errors: the *delta method*.

Effect on $P(\text{foreign})$	weight	mpg
Logit AME	-0.000457	-0.0197
Logit MEM	-0.000518	-0.0224
Probit AME	-0.000465	-0.0207
LPM (robust)	-0.000468	-0.0194

```
. margins, dydx(*)
```

```
Average marginal effects
```

	dy/dx	std. err.	z	P> z
weight	-.0004569	.0000571	-8.01	0.000
mpg	-.0197187	.0096987	-2.03	0.042

```
. margins, dydx(*) atmeans
```

```
Conditional marginal effects
```

```
At: weight = 3019.459 (mean), mpg = 21.2973 (mean)
```

	dy/dx	std. err.	z	P> z
weight	-.0005179	.0001389	-3.73	0.000
mpg	-.0223512	.0127037	-1.76	0.079

Read it: 100 extra pounds lower the probability of being an import by about 4.6 points, on average. And the last two rows of the table: probit AME and LPM land in the same place — the “fair comparison” claim, verified.

4. Effects at chosen values: margins, at() and marginsplot

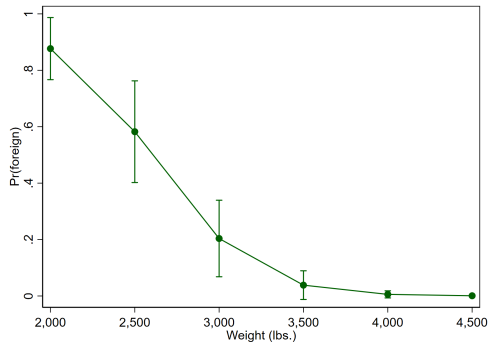
```
. margins, at(weight=(2000(500)4500))
```

```
Predictive margins
```

	Margin	std. err.	z	P> z
weight=2000	.8768344	.0561661	15.61	0.000
weight=2500	.5823404	.0919699	6.33	0.000
weight=3000	.2038047	.0692124	2.94	0.003
weight=3500	.0384522	.0259707	1.48	0.139
weight=4000	.005745	.0064102	0.90	0.370
weight=4500	.000821	.0013048	0.63	0.529

```
. marginsplot
```

- A 2,000-lb car: 88% chance of being an import.
3,000 lbs: 20%. 4,000 lbs: under 1%.
- From 2,500 to 3,000 lbs the probability drops 38 points; from 4,000 to 4,500, half a point. **That** is the non-constant marginal effect — and at() lets you evaluate any scenario, not just the mean.



marginsplot: predicted probability with 95% confidence bands.

5. Factor variables, interactions, and what margins computes

Notation

- `i.sex` categorical, `c.age` continuous, `#` interaction, `##` main effects plus interaction.
- For a dummy declared with `i.`, margins reports the **discrete change** $F(X\beta \mid d=1) - F(X\beta \mid d=0)$, not a derivative.

Two kinds of margin

- *Conditional*: every regressor fixed (e.g. `atmeans`).
- *Predictive*: named regressors fixed, the rest as in the data, then averaged (the default).

Practicalities: empty cells in `sex#group` make margins *not estimable*.

```
. logit foreign c.weight##c.mpg
```

foreign	Coefficient	Std. err.	z	P> z
-----+-----				
weight	-.0010199	.0024909	-0.41	0.682
mpg	.1712339	.2994853	0.57	0.567
c.weight#				
c.mpg	-.0001637	.00014	-1.17	0.242
_cons	7.829103	6.332902	1.24	0.216

```
. margins, dydx(weight) at(mpg=(15 20 25 30))
```

	dy/dx	std. err.	z	P> z
-----+-----				
mpg=15	-.0003716	.0000202	-18.35	0.000
mpg=20	-.000351	2.92e-06	-120.26	0.000
mpg=25	-.0003605	9.97e-06	-36.16	0.000
mpg=30	-.0003897	.0000466	-8.36	0.000

With an interaction the individual coefficients lose their meaning (all “insignificant” here), but the **effect of weight at each mpg** is precise. Always go through margins.

6. How good is the model? (a) The classification table

Predict $\hat{Y} = 1$ if $\hat{P} \geq c$ (default cutoff $c = 0.5$), then cross-tabulate against the truth:

	Predicted 1	Predicted 0
Actual 1	TP (true positive)	FN (false negative)
Actual 0	FP (false positive)	TN (true negative)

- **Sensitivity** = $TP / (TP + FN)$: of the real imports, how many did we catch?
- **Specificity** = $TN / (TN + FP)$: of the real domestics, how many did we leave alone?
- **Accuracy** = $(TP + TN) / N$.

```
. estat classification
```

Classified	True		Total
	D	~D	
+	13	6	19
-	9	46	55
Total	22	52	74

```
Classified + if predicted Pr(D) >= .5
```

Sensitivity	Pr(+ D)	59.09%
Specificity	Pr(- ~D)	88.46%
Positive predictive value	Pr(D +)	68.42%
Negative predictive value	Pr(~D -)	83.64%
Correctly classified		79.73%

80% accuracy sounds good — but calling every car “domestic” already scores 70%. The model catches only 13 of the 22 imports.

6. How good is the model? (b) The cutoff trade-off

Cutoff c	Rule	Sens.	Spec.
1.0	predict 0 for everyone	0	1
0.8	only $\hat{P} > 0.8$ is a 1	low	high
0.5	the usual default	medium	medium
0.3	more generous with 1s	high	low
0.0	predict 1 for everyone	1	0

- Lowering c means more predicted 1s: sensitivity $\uparrow$, specificity $\downarrow$. No free lunch.
- The right c depends on the **cost of each mistake**: missing a fraudulent loan is not the same as annoying an honest customer.
- For auto, $c = 0.3$ catches 18 of 22 imports at the price of 12 false alarms.

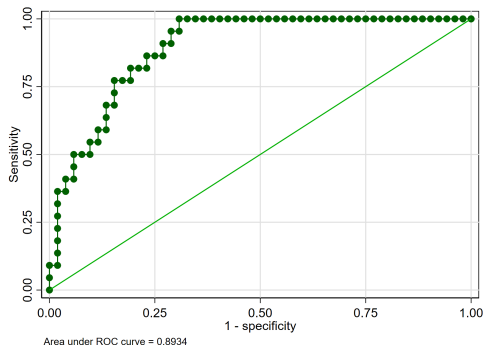
```
. estat classification, cutoff(0.3)
```

Classified	True		Total
	D	~D	
+	18	12	30
-	4	40	44
Total	22	52	74

```
Classified + if predicted Pr(D) >= .3
```

Sensitivity	Pr(+ D)	81.82%
Specificity	Pr(- ~D)	76.92%
Correctly classified		78.38%

6. How good is the model? (c) ROC curve, AUC, and calibration



`lroc:` sensitivity against 1—specificity for every cutoff from 1 down to 0. Area under the curve = 0.893.

ROC traces the whole trade-off of the previous slide in one curve. **AUC** squeezes it into one number: the probability that a randomly chosen import gets a higher $\hat{P}$ than a randomly chosen domestic car.

AUC	Discrimination
0.50	none (a coin toss)
0.60–0.70	poor
0.70–0.80	acceptable
0.80–0.90	good ← auto: 0.89
0.90–1.00	excellent

Calibration is a different question: are the predicted probabilities *right*, not just well ordered? `estat gof, group(10)` (Hosmer–Lemeshow): $\chi^2(8) = 2.30$, $p = 0.97$ — no evidence of misfit.

7. When things go wrong

```
. logit foreign i.rep78
```

```
note: 1.rep78 != 0 predicts failure perfectly;
```

```
1.rep78 omitted and 2 obs not used.
```

```
note: 2.rep78 != 0 predicts failure perfectly;
```

```
2.rep78 omitted and 8 obs not used.
```

```
note: 5.rep78 omitted because of collinearity.
```

```
Logistic regression                Number of obs =      59
```

foreign	Coefficient	Std. err.	z	P> z
-----+-----				
rep78				
1		0 (empty)		
2		0 (empty)		
3	-3.701302	.9906975	-3.74	0.000
4	-1.504077	.9128709	-1.65	0.099
5		0 (omitted)		
_cons	1.504077	.781736	1.92	0.054

Perfect prediction (separation)

- Every car with repair record 1 or 2 is domestic. The ML estimate is $-\infty$; Stata drops the category and its observations. Read the notes.
- Fix: merge sparse categories, drop the variable, or use a penalised logit (firthlogit).

Collinearity: as in OLS, a redundant variable is omitted (here 5.rep78, once 1 and 2 are gone).

Non-convergence: “not concave”, “backed up”, “convergence not achieved” — too many parameters for the data, a near-constant regressor, a scale problem. Simplify; check summarize; rescale.

8. Standard errors: robust, clustered, survey

Stata

```
logit foreign weight mpg, vce(robust)
logit foreign weight mpg, vce(cluster rep78)

svyset psu [pweight=w], strata(strata)
svy: logit y x1 x2
```

- `vce(robust)`: harmless and standard, though a weaker fix than in OLS — if a logit is misspecified, the point estimates are off too.
- `vce(cluster var)`: observations within a village, firm or school are correlated. Needs enough clusters: 5 (here) is far too few; 30–50 is a common rule of thumb.
- `svy`: the weights, strata and PSUs of a complex survey (Susenas, Sakernas, IFLS, DHS). Without it the SEs are wrong.

Std. err. of mpg	default	robust	cluster
value	0.092	0.112	0.027
<i>z</i>	−1.83	−1.51	−5.62
<i>p</i>	0.067	0.131	0.000

Same coefficient (−0.169) in all three columns. The clustered SE is a warning, not a result: with 5 clusters the asymptotics behind it have not begun to work. Never pick the column with the smallest *p*.

Beyond the binary logit

Situation	Stata	Idea
Three or more unordered choices	<code>mlogit</code>	McFadden's model for J alternatives (next lecture)
Ordered categories	<code>ologit</code> , <code>oprobit</code>	One latent index, several thresholds (next lecture)
Panel data	<code>xtlogit</code>	Fixed or random effects for repeated observations
Clustered / multilevel data	<code>meologit</code>	Random intercepts for villages, schools, firms
Rare events, asymmetric link	<code>cloglog</code>	Complementary log-log
Many candidate regressors	<code>lasso logit</code>	Penalised selection
Perfect prediction	<code>firthlogit</code>	Penalised likelihood (user-written)

All of them are the same S-curve with more structure around it. If you understand `logit` and `margins`, you understand most of each.

Hands-on exercise

Dataset: sysuse auto. Add length to the model.

Tasks

- 1 logit foreign weight mpg length — which signs change, which *p*-values? Why might length and weight fight each other?
- 2 logistic ... — report the odds ratio of length per 10 inches.
- 3 margins, dydx(*) — which variable has the largest effect on the *probability*? Compare with the LPM.
- 4 margins, at(length=(140(20)240)) and marginsplot.
- 5 estat classification at cutoffs 0.5 and 0.3; lroc. Did AUC improve on 0.893?
- 6 Re-run with probit. Confirm the coefficient ratio is about 1.6–1.7 and the AMEs are nearly identical.

For discussion: the model classifies the 1978 Buick Opel as an import. A mistake of the model, or of the label?

Common mistakes

- 1 Reading a logit coefficient as a change in probability. It is a change in log-odds; use margins.
- 2 Reading an odds ratio as a probability ratio (“twice as likely”).
- 3 Comparing raw coefficients across logit, probit and LPM — or across samples. Compare AMEs.
- 4 Ignoring the “predicts failure perfectly” notes.
- 5 Reporting accuracy alone when the outcome is rare: 95% accuracy is trivial if 95% are zeros. Report sensitivity, specificity, AUC.
- 6 Too many dummies for a small sample: separation and huge standard errors.
- 7 Clustered standard errors with a handful of clusters.
- 8 Forgetting svy: on survey data.
- 9 Dropping the LPM out of principle. In an experiment, or for a quick look at direction and significance, it is fine — with robust.

Eight stories, eight lessons

Bliss	Zero-one data form an S-curve. Behind it is a latent variable with a threshold.
70% rain	The LPM is real and useful; its flaws (predictions outside $[0, 1]$, heteroskedasticity) are known and tolerated.
Challenger	Fit the curve to <i>all</i> the data. Never select the sample on Y . Extrapolate with your eyes open.
Elo	Probit and logit almost always give the same answer; $\beta_{\text{logit}} \approx 1.6 \beta_{\text{probit}}$.
McFadden	The logit follows from utility maximisation, so it can forecast choices never offered before.
McCleskey	An odds ratio describes a population, not an individual; the controls are never complete.
VioGén	A classifier's errors have victims. Set the cutoff by the cost of each mistake; a score must not replace judgement.
AMS	A coefficient on a group dummy applies to everyone in the group; near a cutoff, a few points decide who gets help.

Next lecture: more than two alternatives — `mlogit` (handwriting recognition) and `oprobit`.

References (1/2): Stories 1–6 and the textbooks

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References (2/2): Stories 7 and 8

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Appendix: the *Challenger* data (Dalal et al. 1989)

Mission	°F	damage	Mission	°F	damage	Mission	°F	damage
STS-1	66	0	41-B	57	1	51-G	70	0
STS-2	70	1	41-C	63	1	51-F	81	0
STS-3	69	0	41-D	70	1	51-I	76	0
STS-5	68	0	41-G	78	0	51-J	79	0
STS-6	67	0	51-A	67	0	61-A	75	1
STS-7	72	0	51-C	53	1	61-B	76	0
STS-8	73	0	51-D	67	0	61-C	58	1
STS-9	70	0	51-B	75	0	51-L	31	?

damage = 1 if there was *thermal distress* (erosion or blow-by) on a field-joint O-ring. STS-4 is unavailable (the boosters were lost at sea). The “number of incidents” charts on the earlier slides use the number of joints (out of 6) affected.